Student Performance Analysis

Statistical Analysis and Regression using R

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Excutive Summary

This project explores student performance using R through data quality assessment, exploratory data analysis, statistical hypothesis testing, correlation analysis, multiple linear regression, and model diagnostics.

The main outcome is the final mathematics grade (`G3`). The analysis examines differences in academic performance and investigates student characteristics associated with final grades.

The findings are interpreted as statistical associations rather than causal effects.

Project Objectives

The objectives are to: - Asses the structure and quality of the dataset. - Explore the distribution of final mathematics grades. - Investigate grades differences across selected student groups. - Examine the relationship between school absences and final grades. - Fit a multiple linear regression model. - Review model fit and regression diagnostics

Dataset Overview

The dataset is obtained from the [UCI Student Performance dataset] (https://archive.ics.uci.edu/dataset/320/student+performance).

The mathematics dataset contains 395 student records and 33 variables.

Table 1: Dataset Overview

Metric	value
Number of students	395

Metric	value
Number of variables	33
Missing values	0
Duplicate rows	0

Key Variables

Variable	Description
`G3`	Final mathematics grade
`sex`	Student sex
`age`	Student age
`studytime`	Weekly study-time category
`failures`	Number of past class failures
`absences`	Number of school absences

Data Preparation and Quality

The data preparation workflow involved reviewing variable types, checking missing values and duplicates, evaluating potential data-quality issues, and preparing variables for statistical analysis.

The original dataset was preserved. Final grades equal to zero ware retained because the available data alone do not establish that these observations are invalid.

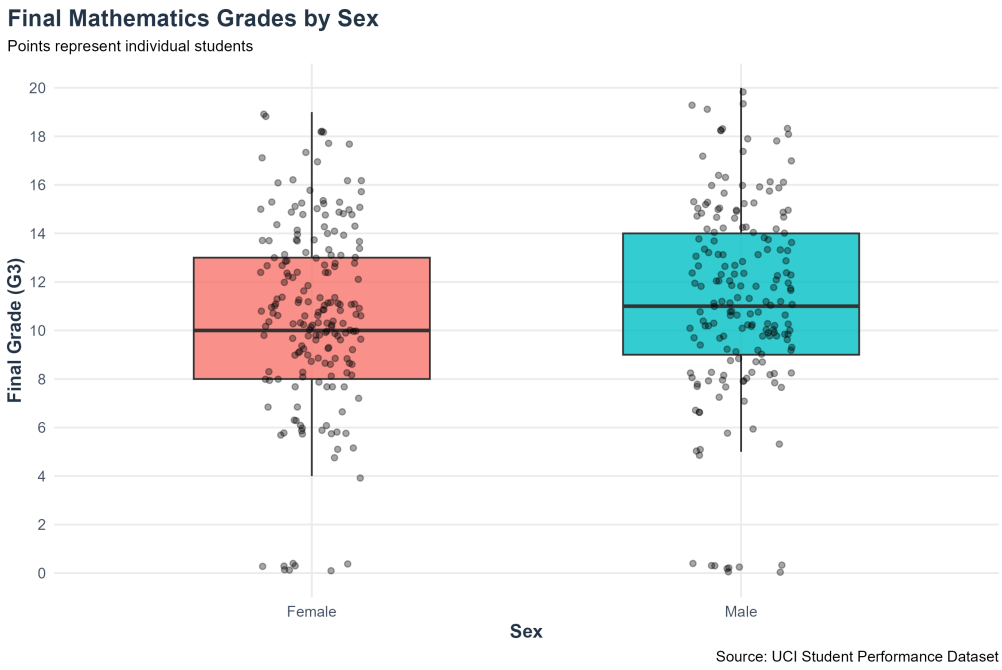
Exploratory Data Analysis

Exploratory data analysis was used to understand the distribution of final grades and explore patterns involving student characteristics.

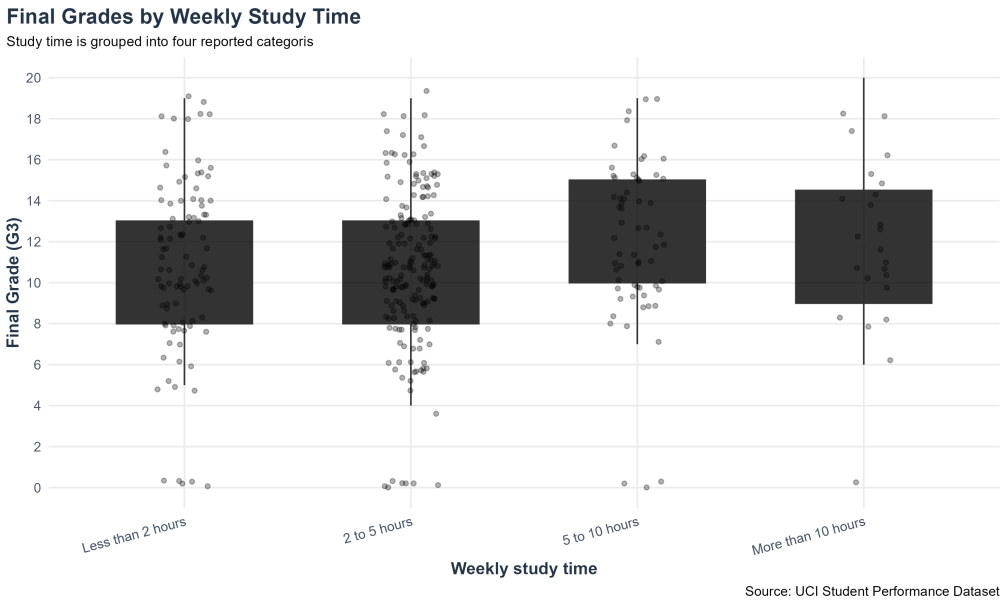
Distribution and Relationships

The following figures were generated during the analysis workflow.

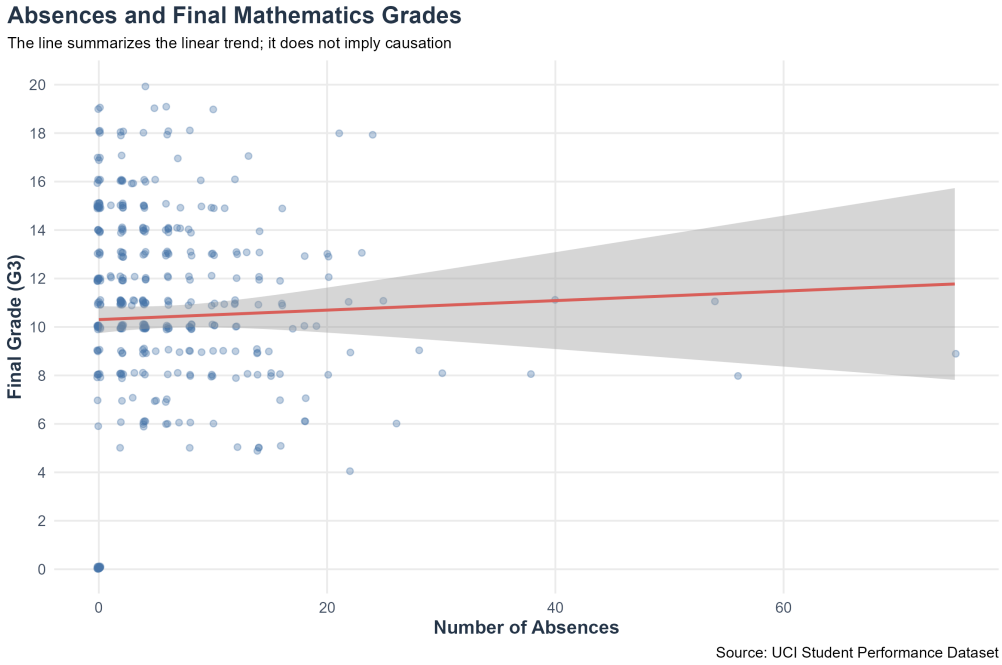
02 Grade By Sex



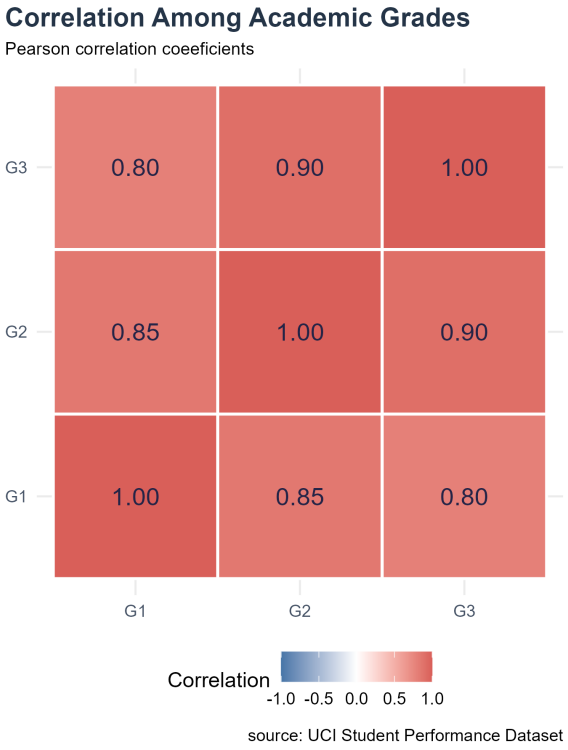
03 Grade By Studytime



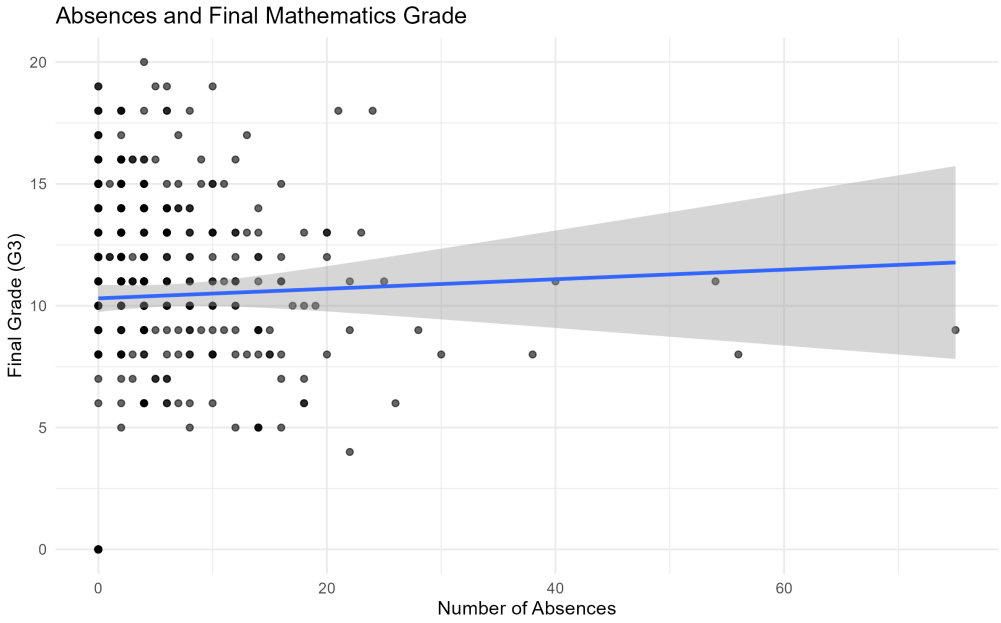
04 Absences Vs Final Grade



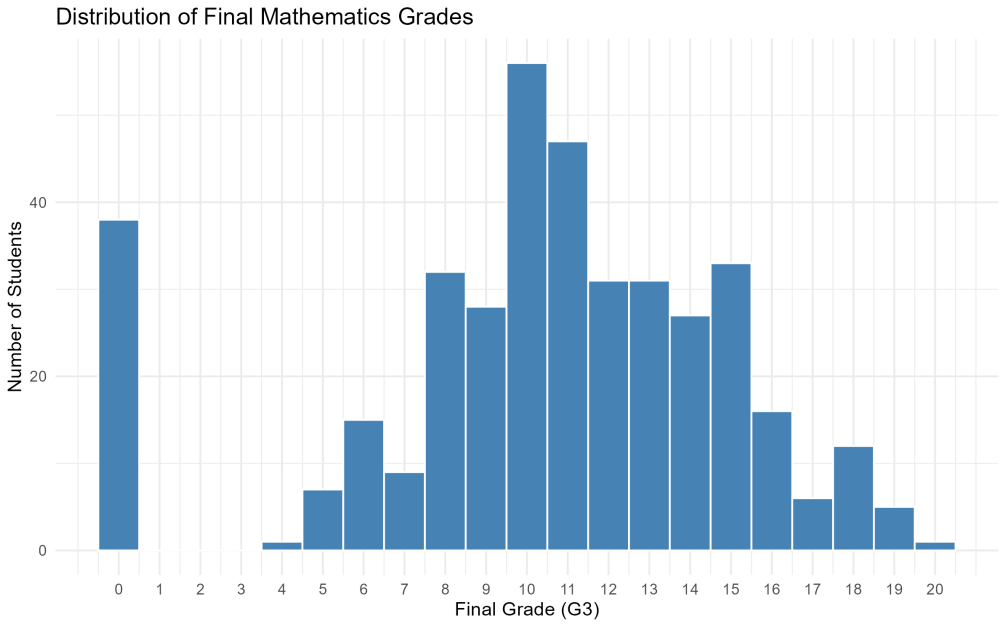
05 Academic Grade Correlation



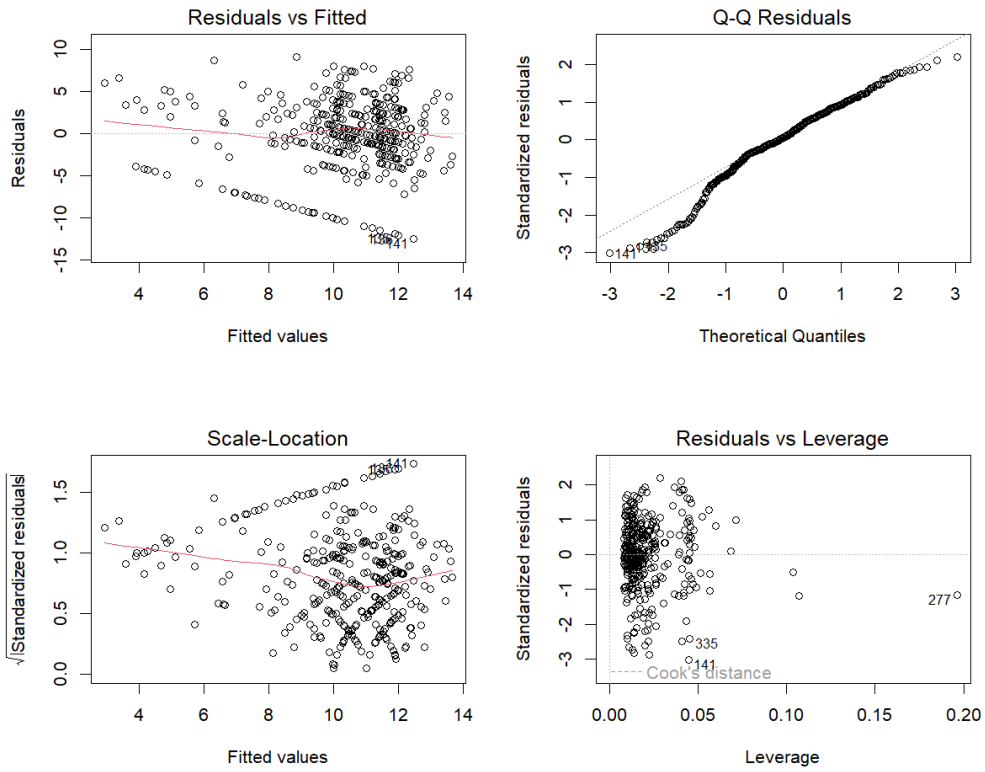
Absences Correlation



Final Grade Distribution



Model Diagnostics



Summary Statistics

Table 2: Summary Statistics for Final Mathematics Grades

Students	Mean	Median	SD	Minimum	Maximum
395	10.42	11	4.58	0	20

Statistical Hypothesis Testing

Three statictical methods were used to investigate the research questions.

Research question	Method
Do final grades differ by sex?	Welch's independent-samples t-test
Do final grades differ across study-time categories?	Kruskal–Wallis test
Is absence count associated with final grades?	Spearman's rank correlation

A 5% significance level was used as a reference for interpretating the results.

Table 3: Hypothesis Testing Results

Research_Question	Method	Statistic	P_Value
Final grades by sex	Welch t-test	-2.0651	0.0396
Final grades across study-time categories	Kruskal-Wallis test	7.5789	0.0556
Absences and final grades	Spearman correlation	0.0177	0.7254

P-values should be interpreted alongside the effect size, the study context, and the assumptions of each method. Statisctial significance does not necessarily imply practical importance.

Correlation and Regression

Baseline Regression Model

A multiple linear regression model was fitted to examine the association between final grades and selected student characteristics.

The model includes sex, age, study time, past, failures, and absences.

Earlier grades (`G1` and `G2`) were excluded from the baseline model because the analysis focuses on student characteristics rather than using previous grades as predictors.

Table 4: Baseline Regression Model Performance

Metric	value
Observations	395.000
R-squared	0.171
Adjusted R-Squared	0.156
Residual standar error	4.210

Regression Coefficients

Table 5: Baseline Regression Coefficients

Predictor	Estimate	Standard_Error	T_Statistic	P_Value
(Intercept)	15.6789	2.9083	5.3910	0.0000
sexM	1.4297	0.4561	3.1347	0.0019
age	-0.3554	0.1751	-2.0294	0.0431
studytime2 to 5 hours	0.1173	0.5323	0.2204	0.8257
studytime5 to 10 hours	1.6627	0.7175	2.3175	0.0210
studytimeMore than 10 hours	0.6972	0.9366	0.7444	0.4571
failures	-2.0714	0.2992	-6.9227	0.0000
absences	0.0534	0.0271	1.9684	0.0497

Each coefficient represents the estimated association with final grade, holding the other variables in the model constant. Categorical coefficients are interpreted relative to their reference categoris.

Model Diagnostics

Regression diagnostics were included in the analysis workflow to review residual patterns and potential issues involving model assumptions.

The diagnostic plots and test outputs should be considered togetger. Teh model is an approximation of student performance, and the results may be affected by unmeasured factors and limitations in the dataset.

Limitations

- The analysis uses observational data and does not establish causality.
- The dataset represent a specific educational context.
- The baseline model explains only part of the variation in final grades.
- Findings depend on the available variables and model assumptions.
- Statistical significance should not be interpreted as proof of practical importance.

Conclusion

This project demonstrates a structured statistical analysis workflow using R, from data preparation and visualization to hypothesis testing, regression modelling, and diagnostic review.

The results provide insight into associations between student characteristics and final mathematics grades while acknowledging the limitations of observational analysis.